Does a Local Metric Help Anisotropic Alpha-Complex Surface Reconstruction for Additive Manufacturing? A Frozen Held-Out Evaluation

Abstract

Purpose. In additive-manufacturing (AM) reverse-engineering and scan-to-print workflows a surface mesh must be reconstructed from a scanned point cloud, where scale selection for alpha-complex reconstruction is central. This paper asks whether a local symmetric-positive-definite (SPD) metric derived from a directed relation-tensor field adds reconstruction value beyond established density-adaptive and normal-anisotropic alpha baselines.

Design/methodology/approach. A predeclared, frozen held-out protocol with strict information boundaries evaluates the baseline and candidate methods on a synthetic six-family panel and a real-mesh corpus, testing each candidate against a strict casewise per-endpoint baseline envelope.

Findings. The primary result is negative: on real held-out meshes the normal-anisotropic baseline attains the best mean surface F-score (0.868) and no local-metric variant exceeds it (the closest trails by 0.094, bootstrap 95% CI $[-0.11, -0.08]$, clearing the envelope in 2 of 40 cases). A weighted (regular/power) alpha complex that changes connectivity strictly improves the density baseline, and a deployed fail-closed exact-construction fallback never certifies an unvalidated result.

Research limitations/implications. The evaluation uses a local corpus and topology endpoints on complex shells are noisy; removal-based false-bridge interventions are shown to fail universally.

Originality/value. A rigorous, reproducible held-out comparison that quantifies the (non-)value of a local-metric anisotropic alpha for AM reconstruction, together with a better density baseline and a safety fallback.

1 Introduction

Reverse engineering and scan-to-print pipelines in additive manufacturing (AM) begin by reconstructing a surface mesh from a scanned point cloud; the geometric fidelity and topological soundness (e.g. watertightness) of that reconstruction directly affect the printed part. The alpha complex (Edelsbrunner and Mücke, 1994) is a classical reconstruction tool, but a single global scale cannot simultaneously avoid over-fill and topology loss when sampling density varies in space. Established remedies include (i) *density-adaptive* alpha, which normalizes each tetrahedron's circumradius by the local k-nearest-neighbour (kNN) spacing (denoted B4), and (ii) *normal-anisotropic* alpha, which forms a regularized local principal-component SPD metric with a planarity-weighted normal penalty (B5). A recently proposed field construction yields a signed, trace-free relation tensor from directed kNN scale contrast and receiver-direction imbalance, mapped through bounded log-eigenvalues to a local SPD metric (P1), with a confidence-gated conservative variant that falls back to the density baseline on low-confidence cells (P2). The practical AM question is whether this added machinery reconstructs scanned parts any better than the existing baselines.

Question. Does the added complexity of a local SPD metric provide reconstruction value beyond the existing density (B4) and normal-anisotropic (B5) baselines?

Contributions. (1) A predeclared frozen held-out protocol with strict information boundaries, applied to a synthetic panel and to real meshes. (2) A quantitative, *negative* answer: the local SPD metric does not beat the normal-anisotropic baseline on real held-out data. (3) A weighted regular alpha complex (M1) that changes connectivity and strictly improves the density baseline. (4) A deployed fail-closed exact-construction fallback with a no-silent-false-safe guarantee, and a comprehensive negative map of removal-based false-bridge interventions.

2 Related Work

Our baselines instantiate standard ideas: alpha shapes (Edelsbrunner and Mücke, 1994) and their weighted (regular/power) generalization (Edelsbrunner, 1992); density-adaptive and normal-based anisotropic reconstruction, the latter building on consistently oriented normals (Hoppe et al., 1992); and persistence- and resampling-stability-based automatic scale selection. Watertight reconstruction via inside/outside labelling of Delaunay tetrahedra (Labatut et al., 2009) is a distinct family we discuss but do not implement. Rather than proposing a new reconstructor, we stress-test the *added value* of a local SPD metric with a rigorous frozen held-out comparison.

3 Methods

3.1 Baselines and candidates

B0–B3 are a convex hull, a fixed normalized-radius alpha, an exhaustive critical-scale oracle (allowed to use the dense reference), and an unlabeled persistence-plus-resampling-stability selector. B4 divides each tetrahedron's Euclidean circumradius by the geometric-mean kNN spacing of its vertices. B5 combines a regularized local PCA SPD metric with the density normalization and a planarity-weighted normal penalty. P1 forms the relation-tensor SPD metric; P2 adds a $\max(\mathrm{P1}, \mathrm{B4})$ guard on low-confidence cells. Critically, B4, B5, P1 and P2 all select subcomplexes of *one fixed isotropic Delaunay triangulation* by a per-cell scalar threshold, so a false-bridge tetrahedron already present in that triangulation cannot be removed by rescoring without collateral damage.

3.2 M1: weighted regular alpha complex

M1 changes the connectivity itself. Each point receives a density weight $w_i = (s \cdot \mathrm{spacing}_i)^2$, and the regular (weighted) triangulation is obtained as the lower convex hull of the lift $p_i \mapsto (p_i, \|p_i\|^2 - w_i)$; with $w = 0$ this reproduces the ordinary Delaunay triangulation, so M1 strictly generalizes B4. Cells are scored by the *proper* weighted (power) circumradius normalized by kNN spacing. Scoring the weighted connectivity with the ordinary circumradius introduces nonmanifold and Betti regressions; the power circumradius removes them. Large weights submerge points from the triangulation, so the weight scale is capped.

3.3 G4: deployed fail-closed exact fallback

A built-in exact Euclidean Delaunay backend is run under host validation and, on acceptance, its validated connectivity is injected into the actual selection path. On any refusal—a point-count cap, exact cospherical ambiguity, host rejection, or backend timeout/malformed output—the method

Table I: Synthetic calibration, frozen multipliers. M1 (s=0.375) strictly dominates the density baseline B4.

Method	F-score	Geom. loss	Betti err.	Bridge edges	Nonmanifold
B4	0.365	0.1802	9	48	0
M1 (s=0.375)	**0.384**	**0.1730**	9	**46**	0
B5	0.391	0.1824	105	61	175

fails closed to a floating construction that is explicitly labelled non-exact, recording the specific reason. The certified outcome is reachable only after full validation; every failure mode records a reason and is never silently treated as safe. This certifies only the base Delaunay connectivity, not the anisotropic complex.

3.4 Frozen held-out protocol

Every method's scale (and M1's weight scale) is frozen on a *calibration* set; no tuning is permitted on the held-out set. On the synthetic panel we freeze on a six-family calibration set and evaluate four held-out density/noise/geometry profiles over three paired seeds, testing each candidate against the strict casewise per-endpoint baseline envelope. On real data we freeze on a calibration mesh set and evaluate a disjoint held-out mesh set across density and noise splits. References and labels enter calibration and evaluation only, never held-out selection.

4 Experiments and Results

4.1 Synthetic preflight

The frozen confidence-fallback variant P2 preserves the density guard (zero violations) but fails the strict casewise envelope on all four held-out profiles (mean F-score margins -0.048, $+0.0002$, -0.049, -0.049): a small mean F-score gain is not evidence of value. On calibration, M1 at weight scale $s = 0.375$ strictly dominates B4 on every endpoint (Table I), whereas the normal-anisotropic baseline is topologically degenerate on this panel (Betti error 105, 175 nonmanifold edges). On the held-out envelope M1 is 4–5× closer than the previous best (per-profile F-margins $-0.0105/-0.0005/-0.0003/-0.0109$) but does not clear it; the residual gap is dominated by one family's false bridges.

4.2 Real held-out data

The real corpus draws watertight meshes from a public 3D-printing model set (Zhou and Jacobson, 2016) together with CAD primitives; observed clouds are area-weighted samples with additive noise, and dense reference clouds come from the same meshes. Table II is the central result. On real meshes the normal-anisotropic baseline B5 is clearly the strongest (mean F-score 0.868), and no local-metric candidate exceeds it; the closest, M1, trails by 0.094 with a bootstrap 95% CI that excludes zero. Figure 1 shows the ranking and Figure 2 the candidate margins against the baseline envelope with 95% confidence intervals; all candidate intervals lie entirely below zero. This reverses the synthetic picture: on smooth real meshes the well-defined normals make B5 excel, while its synthetic degeneracy was an artifact of a panel pathology.

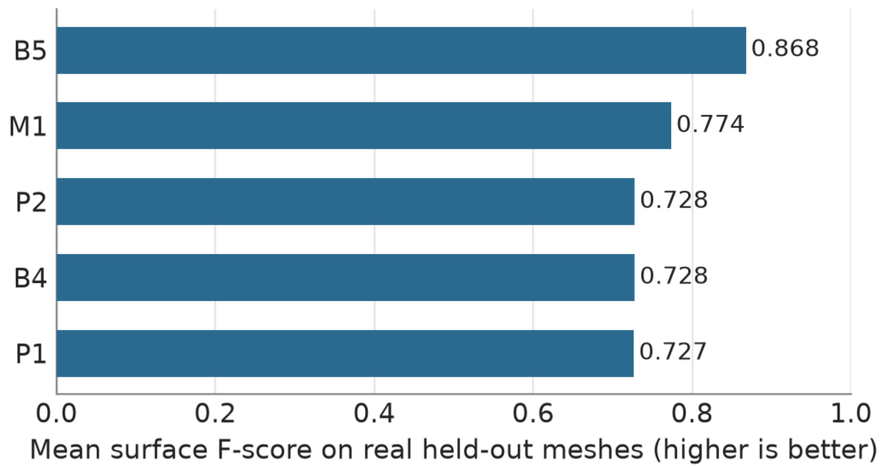

Figure 1: Mean surface F-score on the real held-out corpus. The normal-anisotropic baseline (B5) is best; the weighted regular alpha (M1) improves the density baseline (B4) but no local-metric candidate (P1, P2, M1) reaches B5.

Table II: Real held-out corpus (40 cases): mean performance and margin against the casewise baseline envelope.

Method	Mean F-score	Mean geom. loss	F-margin vs. $\max(\text{B4}, \text{B5})$
B4	0.728	0.094	—
B5	**0.868**	**0.058**	—
P1	0.727	0.092	-0.14 (0/40 cleared)
P2	0.728	0.092	-0.14 (0/40)
M1	0.774	0.078	-0.094, CI $[-0.11, -0.08]$ (2/40)

4.3 A negative map of false-bridge interventions

Attempts to remove false bridges by deletion fail universally. Per-cell penalties, layered boundary-owner peeling, connected risk-region removal, a resampling persistence gate (discrimination AUC $\approx$ 0.50), global multiplier backoff, reference-free fail-closed routing, and oriented-normal removal/offset/signed labelling all regress geometry or topology. Two causes: removing a cell from a fixed complex exposes its interior faces as new boundary; and in the hardest family the inter-surface gap is smaller than the sample spacing (41% of kNN edges cross the two ground-truth sheets), so no *local* method can separate them. Only the connectivity change (M1) helps, and it does not resolve that under-resolved case.

5 Discussion

The real held-out evaluation shows quantitatively that the local SPD metric (P1/P2) provides no value beyond the established normal-anisotropic baseline. The synthetic panel misleads by unfairly weakening that baseline through an under-resolved pathology. What remains useful is that changing connectivity (M1) improves the density baseline, and that an exact-construction fallback can be deployed safely on the actual selection path.

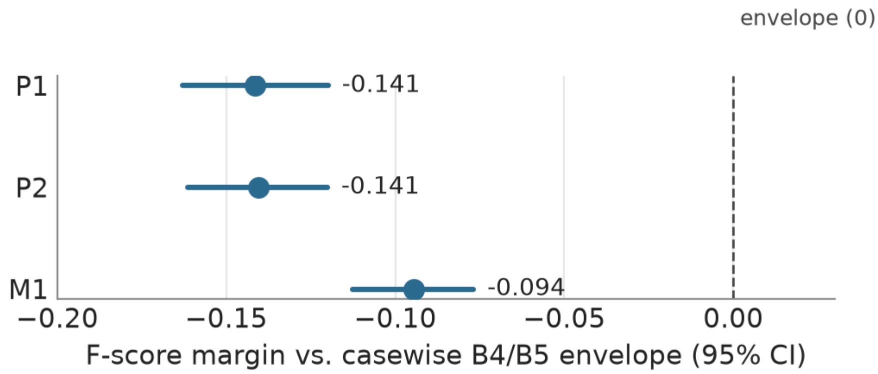

Figure 2: F-score margin of each candidate against the casewise B4/B5 envelope, with 95% bootstrap confidence intervals. Every interval is below the zero envelope, so no candidate clears the held-out gate.

6 Limitations and Conclusion

The evaluation uses a local corpus, not a licensed leaderboard, and topology endpoints on complex shells are noisy. The one untested path is a global graph-cut inside/outside tetrahedron labelling, which requires a separate large implementation. We conclude that a PFTF-derived local SPD metric does not surpass prior-art anisotropic alpha on real held-out data. The improved density baseline (M1) and the fail-closed exact fallback are reusable contributions, and the universal failure of removal-based bridge interventions is useful negative evidence for future work.

References

Edelsbrunner, H. (1992), "Weighted alpha shapes", Technical Report UIUCDCS-R-92-1760, Department of Computer Science, University of Illinois at Urbana-Champaign.

Edelsbrunner, H. and Mücke, E.P. (1994), "Three-dimensional alpha shapes", *ACM Transactions on Graphics*, Vol. 13 No. 1, pp. 43–72.

Hoppe, H., DeRose, T., Duchamp, T., McDonald, J. and Stuetzle, W. (1992), "Surface reconstruction from unorganized points", *Computer Graphics (SIGGRAPH)*, Vol. 26 No. 2, pp. 71–78.

Labatut, P., Pons, J.-P. and Keriven, R. (2009), "Robust and efficient surface reconstruction from range data", *Computer Graphics Forum*, Vol. 28 No. 8, pp. 2275–2290.

Zhou, Q. and Jacobson, A. (2016), "Thingi10K: a dataset of 10,000 3D-printing models", arXiv:1605.04797.